



EPFL MASTER IN FINANCIAL ENGINEERING

Strategic International Asset Allocation using Currency Hedging and Predictive Signals

*An Empirical Study of Diversification, Momentum, and Macro-Finance
Factors*

FIN-405 Investments: Course Project

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1 Introduction

In the context of the Investments course, a hands-on project was conducted to apply concepts of international diversification and dynamic asset allocation. The objective is to construct and analyze portfolios that combine international stock market indices and currency exposures.

By using data spanning from January 2000 to December 2024, we explore how predictive signals (specifically momentum, reversal, carry and dollar) can be beneficial to timing our allocation to international stock markets and currencies in our portfolios. The analysis is conducted using monthly stock return data, exchange rates, and interbank interest rates for seven major international markets: Australia, France, Germany, Japan, Switzerland, the United Kingdom and the United States of America.

Throughout the project, different investment strategies provided in the course are applied, implemented, and analyzed using Python and its data analysis tools. The performance of these strategies is compared, and regressions are run to better understand their underlying drivers and interrelationships. This allows for the assessment not only of which strategies perform best, but also why they work and under what conditions they offer diversification or predictive value.

2 The Data

All necessary data were downloaded from WRDS (country returns) and the FRED website (exchange and interbank rates). After reading all the CSVs and parsing them correctly, the most important and non-trivial processing steps implemented are as follows:

Time-index alignment

- **Equity returns:** Each observation $r_{i,d}$ is re-labelled $r_{i,t}$ with t = last calendar day of the month containing d . This step had to be done since returns were given at the last trading day of the month, which doesn't always coincide with the last calendar day. Aligning to last calendar day makes it easier to match with FX returns.
- **FX levels:** To convert local equity performance to USD, the FX move from the beginning of month t to the end of month $t - 1$ is required. Therefore, every FX series S_t is first aligned to $\text{EOM}(t - 1)$:

$$S_t^{(\text{lag})} = S_{\text{EOM}(t-1)}$$

This step ensures the use of the most relevant exchange rate to convert returns. While a daily exchange rate approach (DEX files from FRED) was considered, it was deemed more appropriate to use an aggregate figure that is temporally close to the return date, minimizing potential imprecision. It is more logical to align FX rates with returns that are at most 2–3 days old rather than with returns that may be up to 30 days delayed. CHF and JPY are also inverted upon loading so that all rates are expressed as “units of foreign per USD.”

- **Sanity filters:** After alignment, months where *either* equity or FX is missing are dropped, guaranteeing a common support for the merge.

Inter-bank (funding) rates

Quarterly 3M inter-bank quotes $R_q^{(3M)}$ arrive as *annual* percentage yields. Before they are merged with the equity panel each observation is converted to its monthly-effective counterpart:

$$R_{\text{monthly}} = \left(1 + \frac{R_{\text{annual}}}{100}\right)^{\frac{1}{12}} - 1.$$

The transformation is applied directly in the data-loader so all downstream objects already reference monthly rates.

Key in-memory structures (excerpt)

Variable	Type	Content / role
returns_dict	dict [str→DataFrame]	Raw local index returns (one per country)
fx_dict	dict [str→Series]	Raw FX spot levels
merged_data	dict	Aligned $\{r_{i,t}, S_{t-1}\}$ pairs
idx_returns	DataFrame	Wide table of USD equity returns
cexr	DataFrame	Wide table of spot FX levels (USD base)
ibexr	DataFrame	3-month inter-bank rates (monthly, %)

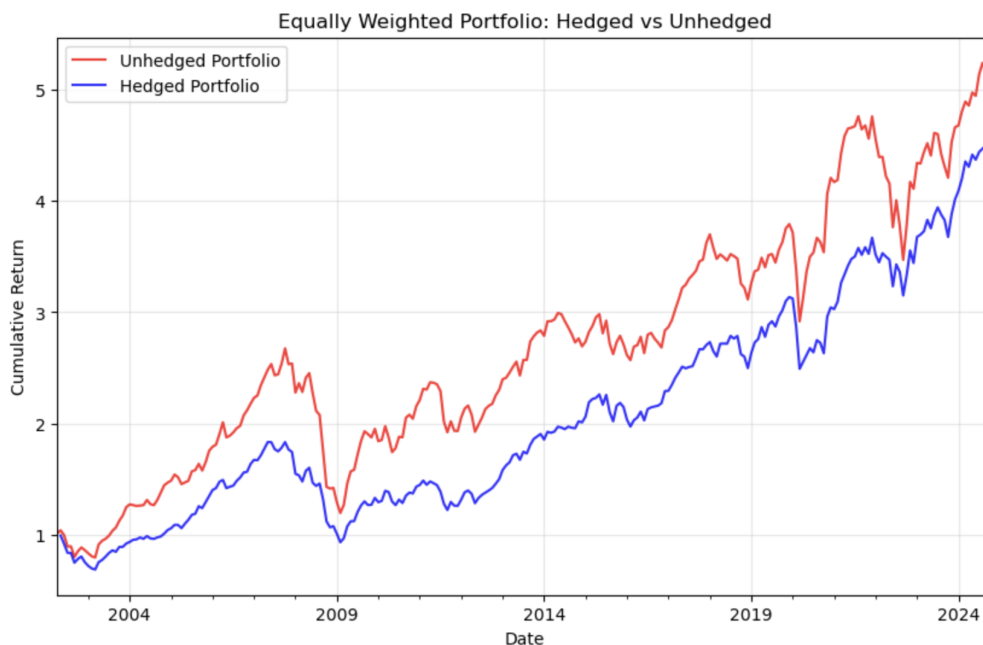
These objects are kept in memory for downstream return-construction steps (hedged & currency-excess) and exported as .csv for reproducibility.

3 The International Diversification Strategy (DIV)

Three alternative multi-asset index portfolios are constructed: Equally-Weighted (EW), Risk Parity (RP), and Mean-Variance Optimised (MVO). Each portfolio is analyzed in both unhedged and currency-hedged form. All portfolios are rebalanced on a monthly basis and evaluated in excess-return space, using the U.S. 3-month Treasury bill as the funding rate. Currency hedging is performed through rolling one-month forward foreign exchange contracts.

Portfolio	Mean Return	Std. Dev.	Sharpe Ratio
Unhedged EW	0.083	0.156	0.521
Hedged EW	0.075	0.131	0.562
Unhedged RP	0.052	0.158	0.319
Hedged RP	0.061	0.132	0.449
Unhedged MVO	0.049	0.187	0.253
Hedged MVO	0.052	0.152	0.334

Table 1 | Performance of the Diversified Strategies (DIV)



Equal Weighted strategy: hedged vs. unhedged

The unhedged portfolio outperforms the hedged one over the long run, mainly due to gains from foreign currency exposure as the USD weakened during key periods. However, hedging smooths returns and reduces volatility by stripping out FX fluctuations. The tradeoff is clear: hedging protects against downside FX risk but can significantly lower total returns in bullish foreign currency environments.

Interpretation.

- Currency hedging lowers volatility across the board ($\approx 20\text{--}25$ bp annualized) while either maintaining or slightly improving mean returns, resulting in uniformly higher Sharpe ratios.
- EW remains the most robust: even after giving up 0.8 % in mean return, its risk compression is large enough to lift the Sharpe from 0.52 to 0.56.

- RP benefits the most in relative terms (Sharpe +13 bps), pushing its risk-adjusted performance close to the hedged EW portfolio.
- MVO is still the weakest of the trio; hedging improves its Sharpe from 0.25 to 0.33, yet the mean-return gain is small.

Overall, hedging acts as a volatility dampener and modest performance enhancer, but portfolio construction choice (EW vs RP vs MVO) retains a first-order impact on risk-adjusted returns

Country	Hedging Pay-off	Visual Take-away
Japan (JPN)	Positive. Hedging outperforms unhedged post-2020 due to yen depreciation. FX-neutral returns are smoother and higher.	Hedged curve ends ~40% above unhedged: $\approx 5.3\times$ vs $\approx 3.7\times$ capital.
Australia (AUS)	Negative. AUD strength and carry made FX exposure beneficial. Hedging lowers return, but also significantly reduces risk and return swings.	Unhedged curve dominates throughout; gap widens post-2016 and persists.

Table 2 | Country-Level Hedging Impact (Summary)

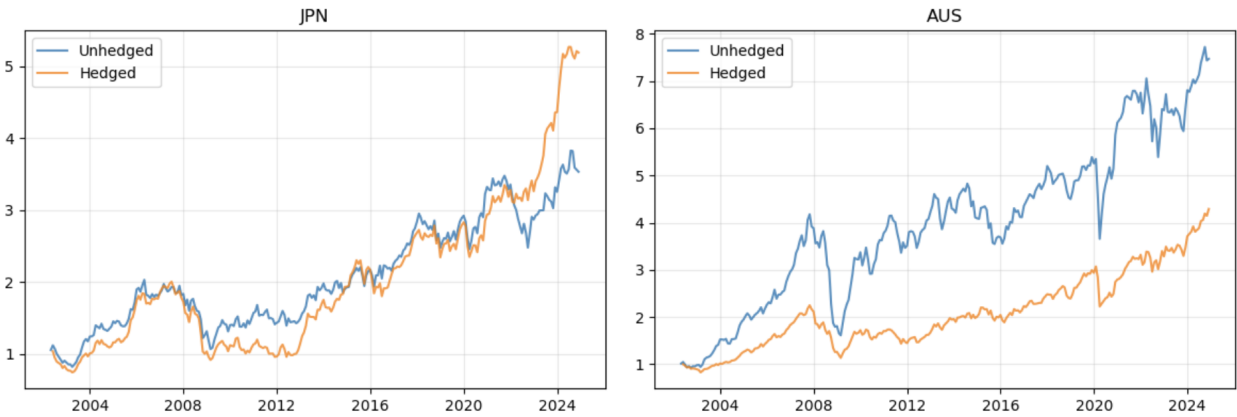


Figure 2 | Currency hedged returns for Japan and Australia

Interpretation:

- **Japan (JPN):** Yen was stable through the 2000s but then weakened sharply after 2012, hedging boosted cumulative returns ($\approx 5.2\times$ vs. $3.7\times$ unhedged by mid-2024).
- **Australia (AUS):** AUD generally appreciated—unhedged returns outpaced hedged ($\approx 7.5\times$ vs. $4\times$), so hedging surrendered substantial FX gains.

Takeaway: Hedging insulates equity returns when the local currency weakens versus the USD but costs you when it strengthens.

4 Equity Index Momentum Strategy (MOM)

A momentum strategy is implemented to examine the extent to which historical returns can predict future performance. On a monthly basis, currency-hedged stock indices are ranked according to their cumulative returns over the preceding 12-month period, excluding the most recent month (i.e., from $t - 12$ to $t - 1$). Let Rank_i^t represent the rank of index i at time t , with higher ranks indicating stronger past performance. Portfolio weights are then assigned to each index based on the following transformation:

$$w_i^t = Z \left(\text{Rank}_i^t - \left(N + \frac{1}{2} \right) \right)$$

where N denotes the number of indices and Z is a normalization factor that ensures the sum of the long weights equals +\$1 and the sum of the short weights equals -\$1. The portfolio is rebalanced dynamically each month using a rolling window of 11 months, which produces updated ranks based on the most recent data. The strategy is computed using the `compute_momentum_returns` function, and the resulting performance is then evaluated using the `analyze_long_short_strategy` function, which yields the following annualized results:

Performance Summary

Portfolio Leg	Mean Return	Std. Dev.	Sharpe Ratio	t-stat (p-value)
Long Leg	0.073	0.135	0.540	2.490 (0.013)
Short Leg	-0.104	0.138	-0.760	-3.510 (0.001)
Total Strategy	-0.031	0.086	-0.360	-1.690 (0.092)

Table 3 | Performance of the Momentum Strategy (MOM)

Interpretation. The long leg of the momentum strategy yields statistically significant positive returns, while the short leg exhibits significant underperformance. This asymmetry reflects that momentum tends to work better when betting on winners than shorting losers, which often revert or face short-selling frictions. The overall strategy produces a small negative return with a Sharpe ratio of -0.36. The t-statistic of -1.69 indicates marginal statistical significance at the 10% level, suggesting limited profitability once both legs are combined.

Regression on DIV Strategy

To examine whether MOM returns are explained by the diversified strategy (DIV), the following OLS regression is run:

$$\begin{aligned} \text{MOM}_t &= \alpha + \beta \cdot \text{DIV}_t + \varepsilon_t \\ \text{MOM}_t &= \underbrace{-0.0019}_{\hat{\alpha}} + \underbrace{0.0051}_{\hat{\beta}} \cdot \text{DIV}_t + \varepsilon_t \end{aligned}$$

	Coefficient	Std. Error	t-statistic	p-value
Intercept (α)	-0.002	0.002	-1.105	0.270
DIV (β)	0.005	0.045	0.114	0.909

Interpretation. The regression reveals no meaningful relation between MOM and DIV. The slope is near zero and statistically insignificant (p-value = 0.909), with an R-squared of 0.000. These findings suggest the momentum strategy delivers returns uncorrelated with the DIV strategy.

5 Equity Index Long-Term Reversal Strategy (REV)

The reversal strategy is designed to investigate potential long-term mean-reversion effects in equity index performance. Each month, currency-hedged stock indices are ranked based on their cumulative returns over a five-year window, excluding the most recent year (i.e., from $t - 60$ to $t - 12$). Let Rank_i^t denote the rank of index i at time t , where higher ranks indicate stronger historical performance. Portfolio weights are computed using the formula:

$$w_i^t = Z \left(\left(N + \frac{1}{2} \right) - \text{Rank}_i^t \right)$$

where N is the number of indices and Z is a scaling constant ensuring that long positions sum to +\$1 and short positions sum to -\$1. This design results in buying long-term underperformers and shorting long-term outperformers, with monthly rebalancing using a rolling window of 48 months. The strategy is computed through the `compute_reversal_returns` function and is further analyzed to produce annual results via the same `analyze_long_short_strategy` routine used for the momentum strategy.

Performance Summary

Portfolio Leg	Mean Return	Std. Dev.	Sharpe Ratio	t-stat (p-value)
Long Leg	0.061	0.134	0.450	1.900 (0.059)
Short Leg	-0.063	0.142	-0.440	-1.850 (0.065)
Total Strategy	-0.002	0.076	-0.020	-0.100 (0.924)

Table 4 | Performance of the Reversal Strategy (REV)

Interpretation. The reversal strategy exhibits marginal yet economically meaningful returns on both legs: the long leg delivers positive returns, while the short leg incurs losses. This outcome suggests that past losers tend to outperform in subsequent periods, consistent with the reversal hypothesis on the long side. However, past winners do not revert as expected; instead, they continue to perform well, reflecting persistent momentum and resulting in losses on the short side. This behavior aligns with the findings from the momentum strategy. Consequently, the combined portfolio yields a near-zero return, indicating that any reversal effect among underperforming indices is offset by the continued strength of past winners.

Regression on DIV Strategy

To evaluate whether the REV strategy's returns are explained by the diversified strategy (DIV), we estimate the following OLS regression model:

$$\text{REV}_t = \alpha + \beta \cdot \text{DIV}_t + \varepsilon_t$$

$$\text{REV}_t = \underbrace{0.0001}_{\hat{\alpha}} + \underbrace{-0.0741}_{\hat{\beta}} \cdot \text{DIV}_t + \varepsilon_t$$

	Coefficient	Std. Error	t-statistic	p-value
Intercept (α)	0.000	0.002	0.080	0.936
DIV (β)	-0.074	0.039	-1.877	0.062

Table 5 | OLS Regression: REV on DIV*R-squared:* 0.017*Observations:* 210*F-statistic:* 3.523

Interpretation. The slope coefficient is marginally significant, suggesting a slight negative co-movement between REV and DIV. However, the R-squared remains low, indicating that the DIV strategy explains little of REV's variation. Despite its lackluster returns, the REV strategy may still offer diversification value in multi-strategy portfolios.

6 Currency Carry Strategy (CARRY)

A currency carry strategy involves borrowing in currencies with relatively low interest rates and investing in currencies with higher interest rates in order to capture the differential, referred to as the "carry." While the strategy has the potential to enhance portfolio returns, it is also exposed to risks, particularly those coming from exchange rate fluctuations.

The strategy is implemented using the U.S. dollar (USD) as the base currency for constructing a long-short portfolio. The interest rate differential (carry) is computed as the spread between the 3-month risk-free interbank rate of each foreign currency and the corresponding 3-month USD rate:

$$\text{Carry}_{i,t} = r_{i,t}^{\text{foreign}} - r_t^{\text{USD}}$$

where i indexes the currency and t denotes the time period in months.

At each month t , currencies are ranked based on their respective carry values, with the highest carry receiving rank N and the lowest receiving rank 1. Currencies with positive carry values are allocated to the long portfolio, while those with negative carry values are allocated to the short portfolio.

Portfolio weights are assigned according to the transformation:

$$w_{i,t} = Z \cdot \left(\text{Rank}_{i,t} - \frac{N+1}{2} \right)$$

where Z is a normalization constant ensuring that the sum of the long weights equals +1 and the sum of the short weights equals -1.

The strategy return at time $t+1$ is given by:

$$R_{t+1}^{\text{CARRY}} = \sum_{i=1}^N w_{i,t} X_{i,t+1}$$

where $X_{i,t+1}$ represents the excess return of a currency-hedged foreign investment in currency i , measured in USD terms.

Performance Summary

Portfolio Leg	Mean Return	Std. Dev.	Sharpe Ratio	t-stat (p-value)
Long Leg	0.0174	0.0792	0.22	1.03 (0.3022)
Short Leg	0.0002	0.0685	0.00	0.01 (0.988)
Total Strategy	0.0172	0.0773	0.23	1.07 (0.2851)

Table 6 | Performance of the Carry Strategy (CARRY)

Interpretation. The carry strategy shows a positive annualized mean return of about 1.72%, mostly driven by the long leg's return. The short leg's return is negligible. However, none of these returns are statistically significant, as their t-statistics are below 1.96 (approximate cutoff for statistical significance at the 5% level). This means we do not have enough evidence to say the return is truly different from zero: the observed return could just be due to randomness or noise.

Regression on DIV Strategy

The following OLS regression model is used to test whether the CARRY strategy's returns are explained by the diversified strategy (DIV):

$$\text{CARRY}_t = \alpha + \beta \cdot \text{DIV}_t + \varepsilon_t$$

$$\text{CARRY}_t = \underbrace{-0.0013}_{\hat{\alpha}} + \underbrace{0.1532}_{\hat{\beta}} \cdot \text{DIV}_t + \varepsilon_t$$

	Coefficient	Std. Error	t-statistic	p-value
Intercept (α)	-0.0013	0.002	-0.802	0.424
DIV (β)	0.1532	0.042	3.686	0.000

Table 7 | OLS Regression: CARRY on DIV

R-squared: 0.061

Observations: 210

F-statistic: 13.58

Interpretation. The regression results show that the CARRY strategy has a statistically significant positive relationship with the DIV strategy, as indicated by the β coefficient of 0.1532. However, the intercept α is not statistically significant, suggesting there is no strong evidence that the CARRY strategy generates excess returns beyond what is explained by DIV. The low value for the R-squared of 0.061 indicates that the variation in the CARRY strategy's returns is not that much explained by movements in the DIV strategy.

7 Currency Dollar Strategy (DOLLAR)

The DOLLAR strategy involves taking a long position in a diversified basket of foreign currencies against the U.S. dollar. The portfolio return is calculated as the average of the excess returns of the foreign currencies versus the dollar:

$$R_{t+1}^{\text{DOLLAR}} = -\frac{1}{N} \sum_{i=1}^N X_{i,t+1}$$

where $X_{i,t+1}$ denotes the excess USD return on a currency-hedged foreign investment.

Performance Summary

Portfolio Leg	Mean Return	Std. Dev.	Sharpe Ratio	t-stat (p-value)
Total Strategy	-0.00426	0.0614	-0.07	-0.33 (0.742)

Table 8 | Performance of the Dollar Strategy (DOLLAR)

Interpretation. The Dollar Strategy shows a small negative annualized mean return of -0.43%, indicating that, on average, the dollar depreciated slightly against the basket of foreign currencies over the sample period. But it is important to note that the p-value is far above the common 5% significance threshold, so the average return is not statistically significantly different from zero: we cannot reject the null hypothesis, and thus the negative performance observed could easily be due to random chance or noise rather than a true underlying pattern.

Regression on DIV Strategy

We use the following OLS regression model to examine if the DOLLAR strategy's returns are explained by the diversified strategy (DIV):

$$\text{DOLLAR}_t = \alpha + \beta \cdot \text{DIV}_t + \varepsilon_t$$

$$\text{DOLLAR}_t = \underbrace{0.0015}_{\hat{\alpha}} + \underbrace{-0.1094}_{\hat{\beta}} \cdot \text{DIV}_t + \varepsilon_t$$

	Coefficient	Std. Error	t-statistic	p-value
Intercept (α)	0.0015	0.001	1.281	0.201
DIV (β)	-0.1094	0.030	-3.63	0.000

Table 9 | OLS Regression: DOLLAR on DIV

R-squared: 0.06

Observations: 210

F-statistic: 13.17

Interpretation. The intercept α is not statistically significant, as indicated by the t-statistic, meaning we cannot reject the null hypothesis that the intercept is zero. The slope coefficient β is statistically significant, indicating a meaningful negative relationship between the DIV and DOLLAR strategies: when the DIV strategy performs well, the DOLLAR strategy tends to perform poorly. However, the R-squared value is low, which implies a weak overall explanatory power despite the statistically significant inverse relationship.

Visual Recap of Core Strategy Behavior

This section provides a concise visual overview of the behavior of the four core strategies analyzed in this report: **Momentum (MOM)**, **Reversal (REV)**, **Carry**, and **Dollar**. Their cumulative returns are compared against the **Diversified (DIV)** benchmark, followed by regression scatterplots showing their relationship to DIV. These visualizations are accompanied by brief interpretations to highlight patterns in performance and co-movement relative to a risk-parity allocation.

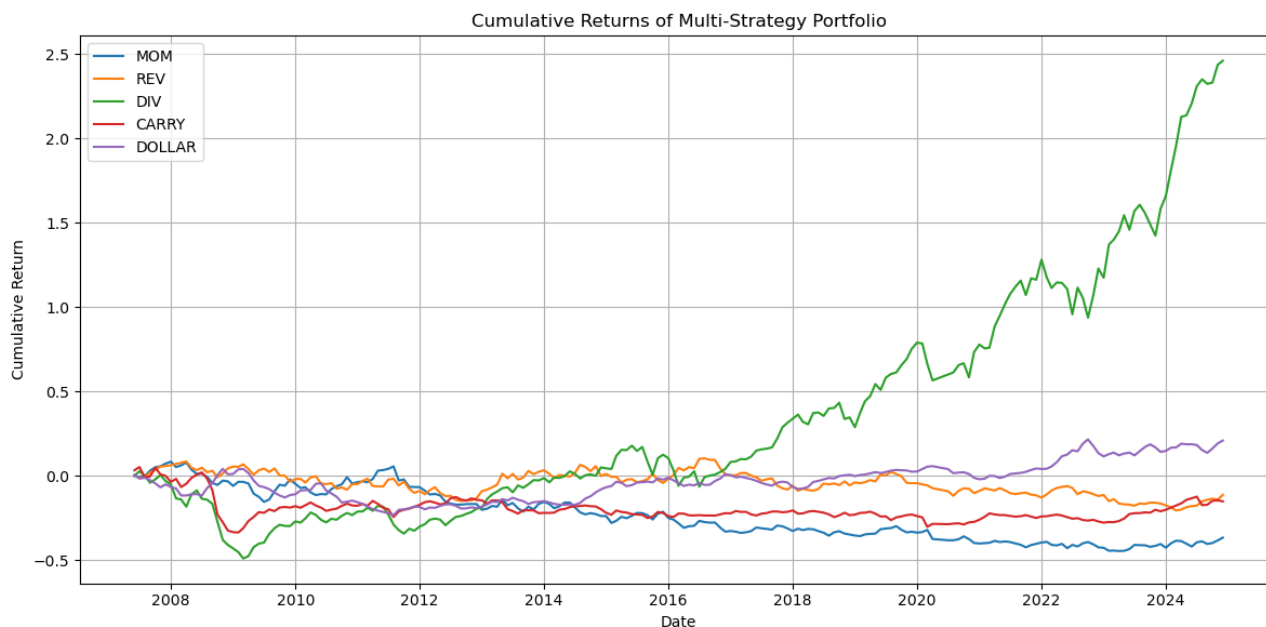
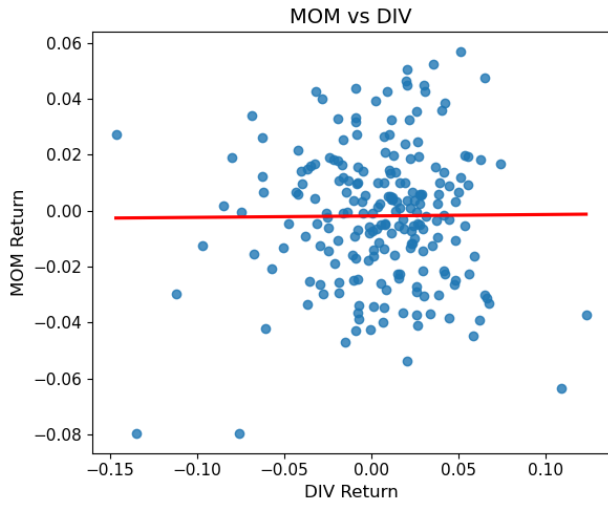
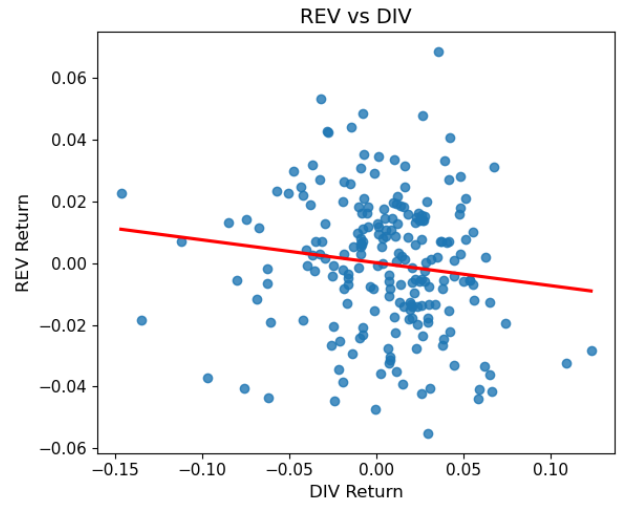


Figure 3 | Cumulative performance of the five core strategies: Momentum (MOM), Reversal (REV), Diversified (DIV), Carry, and Dollar. The DIV strategy shows strong and consistent growth, while MOM and REV underperform.

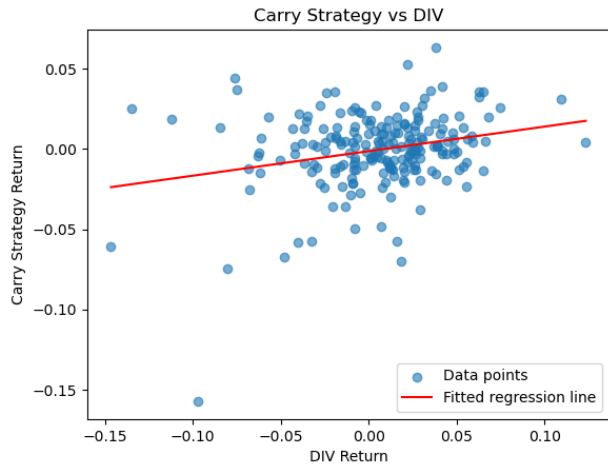
Interpretation. The DIV strategy significantly outperforms the others in terms of cumulative returns. This is intuitive: while strategies like MOM, REV, and Carry may exploit specific informational inefficiencies, they are not designed to optimize for risk-adjusted performance. In contrast, the DIV portfolio is explicitly engineered to deliver balanced exposure and efficient diversification. This superior standalone performance motivates the construction of multi-strategy portfolios, an approach explored in the subsequent section to enhance diversification and mitigate strategy-specific risk.



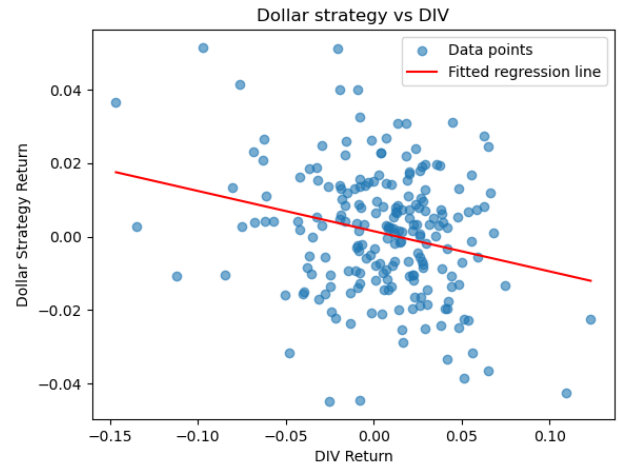
(a) - **MOM vs DIV** Slope: $\hat{\beta} = 0.0051$ ($p = 0.909$); Intercept: $\hat{\alpha} = -0.0019$.



(b) - **REV vs DIV** Slope: $\hat{\beta} = -0.0741$ ($p = 0.062$); Intercept: $\hat{\alpha} = 0.0001$.



(c) - **Carry vs DIV** Slope: $\hat{\beta} = 0.1532$ ($p = 0.000$); Intercept: $\hat{\alpha} = -0.0013$.



(d) - **Dollar vs DIV** Slope: $\hat{\beta} = -0.1094$ ($p = 0.000$); Intercept: $\hat{\alpha} = 0.0015$.

Figure 8 | Regression scatterplots of strategy returns against the DIV benchmark. MOM and REV show weak or marginal relationships, while CARRY and DOLLAR display statistically significant positive and negative slopes, respectively. All intercepts are small and not statistically significant.

Interpretation. While the momentum (MOM) and reversal (REV) strategies exhibit weak or insignificant correlation with the diversified (DIV) portfolio, the carry and dollar strategies show statistically significant relationships, positive for carry and negative for dollar. Across all regressions, the intercepts (α) are close to zero and not statistically significant, indicating no consistent excess return unexplained by DIV. Moreover, the overall explanatory power remains low in all cases, as reflected by modest R-squared values. This outcome aligns with expectations: the DIV strategy is constructed using a risk-parity approach that does not rely on directional return signals such as momentum, reversal, carry, or dollar exposure. The generally low co-movement reinforces the orthogonal nature of these signal-driven strategies relative to a risk-balanced allocation.

8 Optimal Fund Portfolio Return (STRAT)

8.1 Estimating the Optimal Scaling Factor a_t

Consider a fund that allocates capital between the 1-month U.S. Treasury Bill (T-Bill) and a diversified stock index strategy (DIV). The return of the fund at time t is given by:

$$R_{\text{FUND},t} = R_{\text{T-Bill},t} + a_t \cdot (R_{\text{DIV},t} - R_{\text{T-Bill},t})$$

The objective is to determine the time-varying coefficient a_t such that the annualized volatility of the fund remains constant at 15%.

Since T-Bill returns are known in advance (one period ahead) and considered risk-free, the variance of the fund is entirely driven by the excess return component. That is:

$$\text{Var}(R_{\text{FUND}}) = a_t^2 \cdot \text{Var}(R_{\text{DIV}})$$

To ensure the annualized volatility is 15%, the following condition is imposed:

$$\text{Ann. Volatility} = a_t \cdot \sqrt{12 \cdot \text{Var}(R_{\text{DIV}})} = \sqrt{12} \cdot a_t \cdot \text{Std}(R_{\text{DIV}}) = 0.15 \quad \Rightarrow \quad a_t = \frac{0.15}{\sqrt{12} \cdot \text{Std}(R_{\text{DIV}})}$$

The standard deviation of R_{DIV} is estimated using a 60-month rolling standard deviation of the DIV strategy's returns. This yields a time series of dynamic leverage coefficients a_t , which scale the excess return of DIV to maintain the target volatility over time. The resulting evolution is displayed in Figure 9.

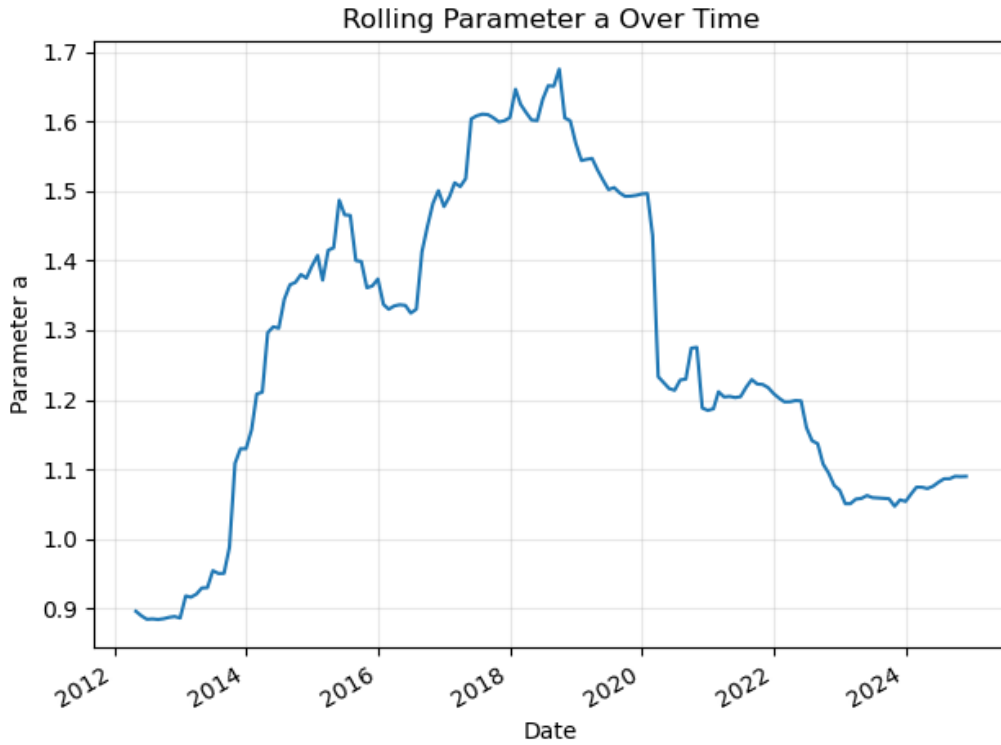


Figure 9 | Evolution of the scaling parameter a_t over time. This parameter dynamically adjusts the exposure to the DIV strategy to maintain a 15% annualized volatility.

8.2 Constructing STRAT via Risk-Parity Allocation

The fund is extended by incorporating an overlay composed of four alternative strategies: two equity-based (MOM and REV) and two currency-based (CARRY and DOLLAR). The objective

is to maintain a target annual volatility of 15% while improving diversification and potentially enhancing returns.

To construct this overlay, denoted R_{STRAT} , a risk-parity framework is used. This approach ensures that each strategy contributes equally to the overall risk of the composite by assigning weights inversely proportional to their estimated volatilities over a 60-month rolling window. Formally, the time-varying weights $w_{i,t}$ are determined, and the STRAT return at time t is computed as follows:

$$R_{\text{STRAT},t} = w_{\text{MOM},t}R_{\text{MOM},t} + w_{\text{REV},t}R_{\text{REV},t} + w_{\text{CARRY},t}R_{\text{CARRY},t} + w_{\text{DOLLAR},t}R_{\text{DOLLAR},t} \quad (13)$$

These weights adjust dynamically in response to changes in each strategy's rolling historical volatility, ensuring a balanced contribution to portfolio risk across all four components. The evolution of weights is shown in Figure 10.

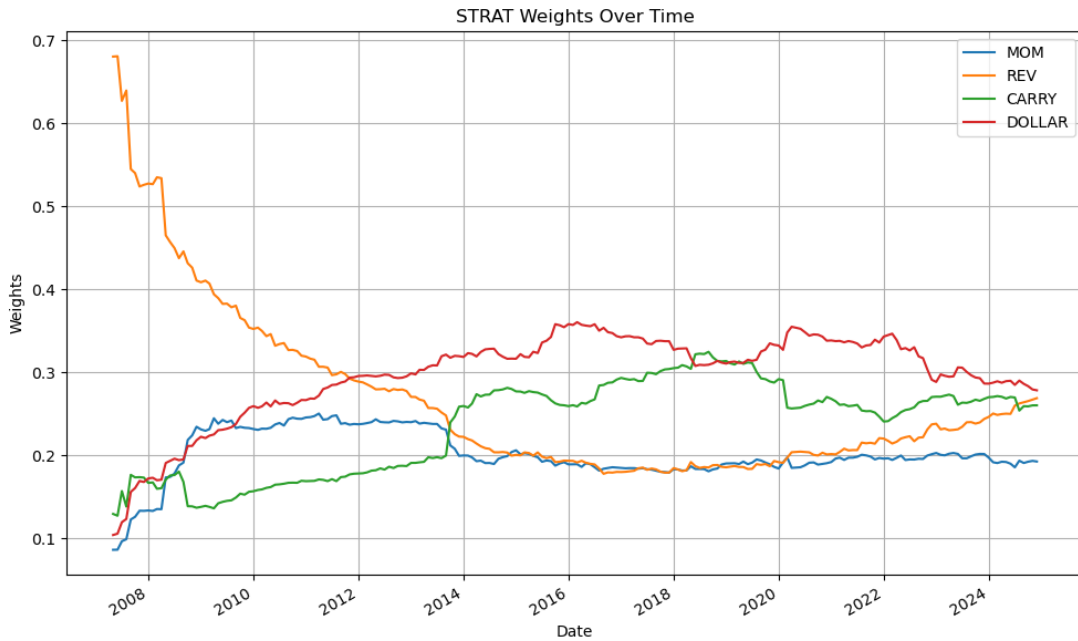


Figure 10 | STRAT Weights Dynamic

The STRAT overlay exhibits a negative average return and Sharpe ratio over the sample period as highlighted in Table 11. This outcome highlights that risk-parity allocation, while effective in balancing volatility contributions, does not guarantee positive performance if the underlying strategies themselves are weak or negatively correlated with returns.

8.3 Mean-Variance Optimization of STRAT

Having constructed the STRAT overlay, the next step is to determine an optimal allocation between the DIV and STRAT strategies that maintain a target annualized volatility of 15% for the overall fund. The resulting strategy is referred to as FUND, and its return at time t is computed as:

$$R_{\text{FUND},t} = R_{\text{T-Bill},t} + b \cdot (R_{\text{DIV},t} - R_{\text{T-Bill},t}) + c \cdot R_{\text{STRAT},t}$$

Here, b and c represent constant portfolio weights applied to the excess return of DIV and the STRAT strategy, respectively. The T-Bill serves as the residual risk-free component. The optimization is subject to the constraint that the resulting portfolio maintains an annual

volatility of exactly 15%. In other words, the following mean-variance optimization problem is solved to determine weights b and c such that the expected return is maximized for a fixed annualized volatility of 15%:

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} \quad \text{subject to} \quad \sqrt{12 \cdot \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}} = \sigma_{\text{target}} = 0.15$$

The optimal solution is given by:

$$\mathbf{w}_t = \frac{\sigma_{\text{target}}}{\sqrt{12 \cdot \sqrt{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}}}} \cdot \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

At each time t , the portfolio weights are allocated as follows:

- DIV: $w_{\text{DIV},t} = b_t$
- T-Bill: $w_{\text{T-Bill},t} = 1 - b_t$
- DOLLAR: $w_{\text{DOLLAR},t} = c_t \cdot w_{\text{DOLLAR}}(R_{\text{STRAT}})$
- CARRY: $w_{\text{CARRY},t} = c_t \cdot w_{\text{CARRY}}(R_{\text{STRAT}})$
- MOM: $w_{\text{MOM},t} = c_t \cdot w_{\text{MOM}}(R_{\text{STRAT}})$
- REV: $w_{\text{REV},t} = c_t \cdot w_{\text{REV}}(R_{\text{STRAT}})$

The weights at each time t are shown in Figure 11, and the average weight allocated to each asset class is reported in Table 10. The FUND strategy consistently allocates positive and relatively stable weights to DIV and T-Bill throughout the sample. This reflects their low volatility and central role in achieving the 15% annualized risk target. In contrast, MOM, REV, CARRY, and DOLLAR receive negative average weights, implying that these components are primarily used as short positions in the optimization process.

A sharp decline in the weights of the overlay strategies is observed in 2021, likely due to increased volatility (Covid-19) or weak return contributions during that period. The optimizer adjusts by reducing exposure to these components in order to maintain the overall fund risk level. This dynamic reallocation highlights the responsiveness of the rolling-window optimization and the importance of adapting to changing market conditions.

	DIV	T-bill	MOM	REV	CARRY	DOLLAR
Mean weight in each strategy	0.91	0.09	-0.49	-0.52	-0.69	-0.83

Table 10 | Mean weights in FUND strategy investment

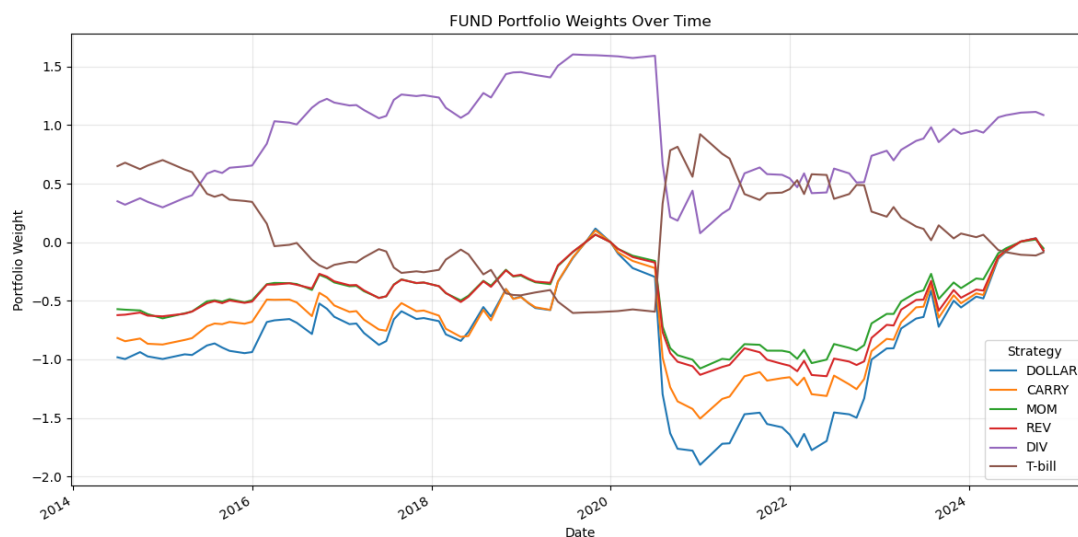


Figure 11 | FUND Strategy Weights Over Time

8.4 Results & Comparisons

A comparison is made between the performance of three versions of the fund strategy:

- **STRAT only:** the risk-parity combination of MOM, REV, CARRY, and DOLLAR.
- **FUND with STRAT:** the mean-variance optimized allocation between T-Bill, DIV, and STRAT, targeting 15% annual volatility.
- **FUND without STRAT:** a scaled allocation between T-Bill and DIV to maintain the same volatility target.

The summary statistics for these strategies are presented in Table 11. While STRAT on its own delivers a negative average return and Sharpe ratio, incorporating it as an overlay improves diversification. The FUND with STRAT achieves a comparable volatility level but a slightly lower Sharpe ratio than the version without STRAT.

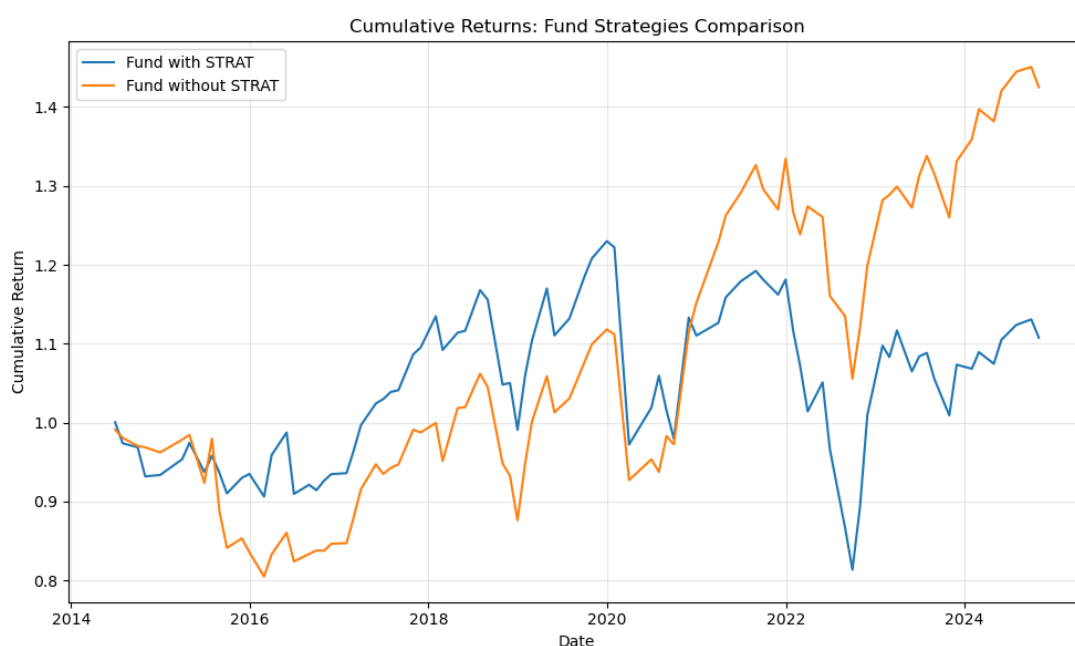


Figure 12 | Comparison of performance of FUND strategies

Strategy	ann. Mean [%]	ann. Std Dev [%]	Sharpe Ratio
STRAT	-0.15	3.56	-0.08
FUND (with STRAT)	8.66	15.80	0.54
FUND (without STRAT)	9.75	15.97	0.60

Table 11 | Summary of performance statistics across fund configurations

As shown in Figure 12, the two fund configurations, those with and without the STRAT overlay, exhibit broadly similar cumulative return patterns over the evaluation period. The trajectories closely mirror one another, indicating that the addition of STRAT neither significantly improved nor impaired short-term performance. However, after 2020, the cumulative return of the fund including STRAT begins to diverge upward, suggesting a stronger contribution from the STRAT overlay during this later phase.

Notably, an investment of \$1 in the fund without STRAT in 2014 would have grown to over \$1.40 by 2025, outperforming the version that included STRAT over the same period (approximately \$1.10).

9 Performance and risk analysis for the Fund strategy

9.1 Regression on Fama-French 5 Research Factors

The Fama-French five-factor model extends the traditional CAPM by including additional sources of systematic risk beyond the market excess return (MKT-RF). These include the size factor (SMB, small minus big), the value factor (HML, high minus low), the profitability factor (RMW, robust minus weak), and the investment factor (CMA, conservative minus aggressive). Together, these factors aim to explain cross-sectional variations in stock returns more effectively by capturing persistent market anomalies related to firm characteristics.

By regressing the monthly returns of both FUND strategies we obtain the following result :

Factor	Fund without STRAT		Fund with STRAT	
	$\hat{\beta}$	<i>p-value</i>	$\hat{\beta}$	<i>p-value</i>
const = α	-0.0014	0.598	-0.0041	0.215
MKT-RF	0.8437	0	0.8292	0
SMB	-0.0452	0.682	-0.1075	0.421
HML	0.1767	0.086	0.4182	0.001
RMW	-0.0014	0.992	0.1017	0.546
CMA	-0.1670	0.307	-0.4108	0.039
R^2	0.717		0.657	

Table 12 | OLS regression results on Fama-French five factors for the FUND strategies with and without STRAT

Fund with STRAT: The regression results show that the excess return on the market (MKT-RF) is statistically significant at the 5% level, with a coefficient of 0.8292 and a p-value of 0. This indicates that the fund's returns are strongly driven by overall market movements. In addition, the investor factor (CMA) and the value factor (HML) are both statistically significant, with p-values of 0.039 and 0.001, respectively. The positive loading on HML indicates a preference for value over growth.

The remaining factors are not statistically significant, providing no strong evidence that they contribute to explaining the fund's returns. The overall R^2 of 0.657 indicates that the five Fama-French factors explain approximately 66% of the variability in the fund's monthly returns.

Fund without STRAT : In the case of the fund without STRAT, the results are similar in that the market factor (MKT-RF) is again highly significant (coefficient = 0.8437, $p = 0$), confirming that market risk remains the primary driver of returns. None of the other four factors are statistically significant at the 5% level, and the intercept (α) also fails to reject the null hypothesis ($p\text{-value} = 0.598$).

The model's R^2 of 0.717 indicates that the five factors account for approximately 72% of the variation in monthly fund returns, a slightly higher explanatory power than in the case with STRAT included.

9.2 EMH, CAPM and APT

EMH The Efficient Market Hypothesis (EMH) suggests that abnormal returns should not be predictable, as prices reflect all available information. In both fund regressions, the intercept (α) is not statistically different from zero, meaning we fail to reject the null hypothesis $\alpha = 0$. This supports the weak-form EMH, implying no abnormal returns beyond those explained by systematic factors.

CAPM According to CAPM, only market risk should be rewarded. In both regressions, the market factor (MKT-RF) is highly significant, confirming it as the primary driver of returns. This aligns with CAPM, suggesting our fund is mainly exposed to systematic market risk. However, when STRAT is included, both CMA and HML also become statistically significant, indicating that the strategy is sensitive to factors that go beyond the single-factor CAPM framework. The R^2 values of 0.717 (without STRAT) and 0.657 (with STRAT) show that a substantial portion of return variation is captured by the model. Overall, while the results are broadly consistent with CAPM, the significance of additional factors points toward a more multifactor explanation of returns.

APT APT allows for multiple factors to influence returns. In our results with STRAT, CMA and HML are statistically significant in addition to MKT-RF, supporting the idea that several risk sources matter. While RMW and SMB remain insignificant, the presence of three significant loadings lends moderate support to APT. These results suggest that multiple systematic exposures drive returns, consistent with the APT framework.